



# West Bengal State Electricity Distribution Company Limited

(A Government of West Bengal Enterprise)

ওয়েস্ট বেঙ্গল স্টেট ইলেক্ট্রিসিটি ডিস্ট্রিবিউশন কোম্পানী লিমিটেড

(পশ্চিমবঙ্গ সরকারের একটি উদ্যোগ)

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Office Order no: 08/2018 dated 12/10/2018

With a view towards minimizing the nos. of Accidents occurring frequently due to unsafe conditions of WBSEDCL's System & unsafe act of Technical Workforce, an Action Plan has been formulated. The Plan comprises of Long Term measures as well as Short Term & Continuous measures for concerned area of System, synopsis of which is detailed below.

## A. LONG TERM MEASURE

| Serial No | Brief Description of the unsafe Condition                 | Target Timeline                                                                                                         |
|-----------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| A         | Conductor snapping & Loose Sag.                           | For survey & estimate: By 31 <sup>st</sup> March'2019<br>Procurement & Implementation: By 31 <sup>st</sup> October'2019 |
| B         | Dual/ multiple source in a single structure.              | For survey & estimate: By 31 <sup>st</sup> March'2019<br>Procurement & Implementation: By 31 <sup>st</sup> October'2019 |
| C         | Broken Kiosk/ unsafe DTR installation.                    | For survey & estimate: By 31 <sup>st</sup> March'2019<br>Procurement & Implementation: By 31 <sup>st</sup> October'2019 |
| D         | Single switchgear operating two or more outgoing feeders. | For survey & estimate: By 31 <sup>st</sup> March'2019<br>Procurement & Implementation: By 31 <sup>st</sup> October'2019 |
| E         | Cradle Guard/ Tie Guard Crossing point                    | For survey & estimate: By 31 <sup>st</sup> March'2019<br>Procurement & Implementation: By 31 <sup>st</sup> October'2019 |

## B. SHORT TERM MEASURE

| Serial No | Brief Description of the unsafe Condition / act:                                                                                                            | Target Timeline                                                                                                          |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| A         | Leaning of poles & lowering of conductor /damaged PCC poles/ existence of wooden poles/ improper earthing of LA at UG Cable                                 | For survey & estimate: By 31 <sup>st</sup> December'2018<br>Procurement & Implementation: By 31 <sup>st</sup> March'2019 |
| B         | Fixing of spacers                                                                                                                                           | For survey & estimate: By 31 <sup>st</sup> December'2018<br>Procurement & Implementation: By 31 <sup>st</sup> March'2019 |
| C         | Diversion of one outgoing feeder to another Substation due to emergency/ load diversion.                                                                    | Implementation by 31 <sup>st</sup> December'2018                                                                         |
| D         | Malfunctioning of 33KV DBCR/ TPGO Isolator at Substation & 11KV Isolator at DTR.                                                                            | For survey & estimate: By 31 <sup>st</sup> January'2019<br>Procurement & Implementation: By 30 <sup>th</sup> April'2019  |
| E         | Missing of feeder identification & non availability of requisite number of Shut down/ Break down/ Feeder diversion boards at some of the 33 KV Substations. | Positively by 31 <sup>st</sup> December'2018                                                                             |
| F         | Non-availability of switching instruction at Substation                                                                                                     | Positively by 31 <sup>st</sup> January'2019                                                                              |
| G         | Monthly Safety Meeting at Division & Zonal Level.                                                                                                           | Immediate Action.                                                                                                        |
| H         | Non availability/ non use of proper PPEs by Technical work force for HT & LT network                                                                        | Immediate Action                                                                                                         |
| I         | Improper/ insufficient documentation at Substation / CCCs.                                                                                                  | Immediate Action                                                                                                         |
| J         | Routine & Surprise visit at worksite during any execution for observance of SoP and availability & use of PPEs.                                             | Immediate Action                                                                                                         |



**C. CONTINUOUS MEASURES FOR GENERATION OF AWARENESS AMONG GENERAL PUBLIC AND TECHNICAL WORKFORCE & SAFETY TRAINING**

| Serial No | Brief Description of the unsafe Condition / act: | Target Timeline                                                                                                                                                                                                                               |
|-----------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A         | Safety Awareness Programme for General Public.   | At least one programme in every month at each Division.                                                                                                                                                                                       |
| B         | Safety Awareness Programme for Technical Workers | At least one bi-monthly programme at each Division.                                                                                                                                                                                           |
| C         | Safety Training Programme for Technical Workers  | In addition to Scheduled Training as per prevailing Training calendar, a special drive of at least one Safety Training to be imparted to those Technical Workers, who are yet to receive any Training so far, by 31 <sup>st</sup> August'2019 |

The detailed Action Plan with time frame & Implementing Authority is appended.

Since the plan is relating to up keeping & maintenance of our distribution network in good condition the main responsibility of implementation of the Safety Action Plan vests with the Distribution wing along with necessary supports from the Project wings. Under this situation Chief Engineer (Distribution) is requested to follow up for its proper implementation on a regular basis out of his existing resources, which may be reviewed subsequently in PRT Meeting.

Besides that Training wing has a very effective part in the Action Plan which may be implemented therein in a rigorous manner.

This is for information & necessary action by all concerned in the Distribution set up. Safety wing is responsible for close follow up of the plan in consultation with the Zonal Managers & Regional Managers on regular basis and provide input/support as may be required from time to time

A quarterly compiled report on the progress of implementation of the Action Plan to be submitted by the Zonal Authority to Corporate Safety Cell with endorsement to the Director: Distribution & Chief Engineer: Distribution for review in the PRT Meeting.

Encls:- As stated

  
(Sujay Sarkar)  
Director: HR

Memo no: Safety/5/2018/ 180 (1-125)  
Distribution:

Dated: 12/10/2018  
TB

1. The Chairman & Managing Director, WBSEDCL.
2. The Director: Distribution/ Finance, WBSEDCL.
3. The Legal Advisor, WBSEDCL.
4. The Chief Engineer: Distribution/ Procurement, WBSEDCL.
5. The General Manager: (HR&A), WBSEDCL.
6. The General Manager: HRD & Training Department, WBSEDCL - for advice to the Principal of all EETI & Training Institute at Kalyani, Nadia for their information & necessary action.
7. The Zonal Manager/ Sr. Manager (HR&A): all Zones, WBSEDCL.
8. The Regional Manager/ HR&A Officer: all Regions, WBSEDCL.
9. The Divisional Manager: all Divisions, WBSEDCL - for guidance & circulation to Assistant Engineer: Technical/ Station Manager/ Assistant Manager (HR&A) under his jurisdiction for immediate action.

**LONG TERM MEASURES**

(involving large quantity of material)

**A. Brief Description of the unsafe condition: Conductor snapping & Loose Sag.****Details of unsafe condition observed--**

- i. Very old, aged & brittle conductor
- ii. Conductor size not commensurate with loading pattern
- iii. Existence of GI wire in HT line
- iv. Multiple joints in a single span & jointing sleeve not used in any joint
- v. Too large Span compared to standard
- vi. Improper Tree trimming, infringement is there due to RoW problem

**Steps for improvement--**

- i, ii & iii. Phase-wise replacement of such conductor & conversion from AAC/ GI to ACSR/ AB Cable/ Insulated conductor as per Site requirement
- iv. Purchase of jointing sleeve at Division level & insisting use by the Agency as & when required.
- v. Use of online pole to reduce span length
- vi. Intensive tree trimming programme after patrolling to be finalised & to be taken up with the concerned Municipal/ Panchayet authority to overcome RoW problem, as & where persists.

**Action to be taken by:**

Survey work with estimate to be done by AE: Tech for HT/ Station Manager for LT system [with direct assistance of SAE(E)]

Procurement action with order for execution by Divisional Manager

**Proposed Time frame:**

For survey & estimate: By 31<sup>st</sup> March'2019

Procurement & Implementation: By 31<sup>st</sup> October'2019

**B. Brief Description of the unsafe condition: Dual/ multiple source in a single structure.****Details of unsafe condition observed:**

- i. HT & LT source in a single structure
- ii. Single structure with dual LT feed from two different DTRs.
- iii. Dual HT source in a single structure
- iv. Single structure with multiple DTR and isolator
- v. 11KV Changeover System on multi pole structure with Isolators at 33KV Substation location

**Steps for improvement--**

- i. Survey is required to find out the exact location against each feeder emanating from 33/11 KV Substation, provision for increasing of clearance & fixing of cradle guard to be made.
- ii. Survey is required to find out exact location in each CCC. Box bracket with appropriate size to be fixed to separate the source with proper signage.
- iii. Survey is required to find out exact location under each 33/11 KV Substation, provision for dead span to be made with proper signage.



- iv. Survey is required to find out exact location under each CCC. Provision of proper signage to be arranged.
- v. Survey is required to find out exact location under each 33KV Substation. Provision of proper signage to be arranged.

**Action to be taken by:**

Survey work with estimate to be done by AE: Tech for HT/ Station Manager for LT system [with direct assistance of SAE(E)]

Procurement action with order for execution by Divisional Manager

**Proposed Time frame:**

For survey & estimate: By 31<sup>st</sup> March'2019

Procurement & Implementation: By 31<sup>st</sup> October'2019

**C. Brief Description of the unsafe condition: Broken Kiosk/ unsafe DTR installation**

**Details of unsafe condition observed:**

- i. Broken Kiosk door/ Kiosk is not installed at DTR structure.
- ii. Isolator handle is broken at DTR installation.
- iii. Rated fuse is not installed at HT/ LT side at DTR structure.
- iv. Common earthing arrangement at DTR neutral and Structure.
- v. Lower height of DTR/ installation of DTR at man height with Isolator & fuse unit
- vi. Lower gap between Isolator male female contact and Isolator & DTR

**Steps for improvement--**

- i. Survey is required for no of DTR without Kiosk/ broken Kiosk door in each CCCs, replacement is required phase wise
- ii. Survey is required for locating the broken isolator in each CCC, replacement to be done on urgent basis.
- iii. Routine survey is required for locating the non-standard gauges of fuse used in LT Kiosk in each CCCs with instant replacement
- iv. Survey is required for locating poor earthing arrangement for transformer neutral and iron structure. Rectification to be done phase wise.
- v. Survey is required for locating DTR installed at lower height. Action for placement at proper height with installation of Isolator with fuse unit to be taken & if increasing of height is not feasible due to site constraint, permanent barricading with installation of danger board to be arranged.
- Vi. After survey, action for maintaining proper gap between Isolator male female contact and Isolator & DTR to be arranged .

**Action to be taken by:**

Survey work with estimate to be done by AE: Tech for HT/ Station Manager for LT system [with direct assistance of SAE(E)]

Procurement action with order for execution by Divisional Manager

**Proposed time frame:**

For survey & estimate: By 31<sup>st</sup> March'2019

Procurement & Implementation: By 31<sup>st</sup> October'2019



**D. Brief Description of the unsafe condition: Single switchgear operating two or more outgoing feeders.**

**Details of unsafe condition observed:**

In absence/ breakdown of one outgoing breaker, two or more different feeder is controlled through one, resulting to mistake during shutdown from Substation.

**Steps for improvement:**

The problem to be sorted out immediately after any such breakdown for repair/ replacement of the defective gear. In case of non-existence of any breaker against any particular feeder in the original scheme, proposal to be initiated for early installation of new breaker

**Action to be taken by--**

Survey work with estimate to be done by AE: Tech for HT/ Station Manager for LT system [with direct assistance of SAE(E)]

Procurement action with order for execution by Divisional Manager

**Proposed Time frame:**

For survey & estimate: By 31<sup>st</sup> March'2019

Procurement & Implementation: By 31<sup>st</sup> October'2019

**E. Brief Description of the unsafe condition: Cradle Guard/ Tie Guard at Crossing point**

**Details of unsafe condition observed:**

- i. Damaged or missing Cradle guards/ insufficient clearance at HT-HT or HT-LT Line crossing/ road crossing/ river or canal crossing/ market place/ school compound/ congested area
- ii. Damaged/ missing tie guard between EHV & HV feeder crossings

**Steps for improvement:**

- i. Survey is required for identification of exact location of the strategic places for maintaining of proper clearance with repairing/ new fixing of cradle guards as & where necessary.
- ii. Survey is required for identification of exact location for crossing point of EHV & HV feeder, rectification to be done on urgent basis

**Action to be taken by:**

Survey work with estimate to be done by AE: Tech for HT/ Station Manager for LT system [with direct assistance of SAE(E)]

Procurement action with order for execution by Divisional Manager

**Proposed time frame:**

For survey & estimate: By 31<sup>st</sup> March'2019

Procurement & Implementation: By 31<sup>st</sup> October'2019



### **SHORT TERM MEASURES**

involving minimal quantity of material

#### **A. Brief Description of the unsafe condition: Leaning of poles & lowering of conductor/ damaged PCC poles/ existence of wooden poles/ improper earthing of LA at UG Cable**

##### **Details of unsafe condition observed:**

- i. Uprooting of stay wires
- ii. Wooden poles for HT/ LT lines
- iii. PCC poles with wooden bracket
- iv. Worn out diagonals and cross bracings in HT DP/TP/ 4 pole structure
- v. Leaning of poles
- vi. Absence of Anti climbing devices/ Danger board at poles
- vii. Worn out/ missing earthing of Cross arms, insulator pins etc of PCC pole
- viii. At HT UG Cable structure, separate earth electrode is not used for LA earthing

##### **Steps for improvement:**

- i. Survey is required for locating the uprooted stays for HT & LT network. After survey, reinstallation of stay/ use of strut pole & re-sagging as & where necessary to be done.
- ii. Survey is required for identification the wooden poles at HT/ LT network, urgent replacement is required.
- iii. Survey is required for locating the PCC poles with wooden brackets, replacement is required in urgent manner
- iv. Survey is required to locate the worn out iron fittings and diagonals in HT structure, urgent replacement is required.
- v. Survey is required to locate the leaning poles and corrective measure is required on urgent basis
- vi. Survey should be carried out for each feeder, reinstallation to be done through daily maintenance work
- vii. Survey to be conducted during patrolling to identify the locations & corrective measure to be taken on urgent basis.
- viii. Survey to be conducted during patrolling to identify the locations & corrective measure to be taken on urgent basis.

##### **Action to be taken by:**

Survey work with estimate to be done by AE: Tech for HT/ Station Manager for LT system [with direct assistance of SAE(E)]

Procurement action with order for execution by Divisional Manager

##### **Proposed Time frame:**

For survey & estimate: by 31<sup>st</sup> December'2018

Procurement & Implementation by 31<sup>st</sup> March'2019

#### **B. Brief Description of the unsafe condition: Fixing of spacers**



**Details of unsafe condition observed:**

- i. Spacer is damaged/ missing in location like market place/ across the road/ school compound etc.
- ii. Inadequate/ excess spacer is installed some of the LT over head lines

**Steps for improvement:**

- i. Survey with replacement/ new installation of requisite spacer at LT over head line as a routine job.
- ii. Monitoring is required during installation of spacer at LT over head line & installation of additional spacers/ dismantling of excess spacers to reduce unnecessary sagging of conductor.

**Action to be taken by:**

Survey work with estimate to be done by Station Manager [with direct assistance of SAE(E)]  
Procurement action with order for execution by Divisional Manager

**Proposed time frame:**

For survey & estimate: by 31<sup>st</sup> December'2018  
Procurement & Implementation by 31<sup>st</sup> March'2019

**C. Brief Description of the unsafe condition:** Diversion of one outgoing feeder to another Substation due to emergency/ load diversion.

**Details of unsafe condition observed:**

During breakdown of any line/ Breaker/ PTR, sometimes it becomes necessary to temporarily divert any incoming or outgoing feeder to any other breaker or any other source. It has been observed that due to miscommunication, accident occurred due to improper shutdown for non-availability of information to Operators of adjacent 33KV Substation/ executing Agency.

**Steps for improvement--**

To avoid accident due to this aspect, proper communication with signage at both the original & diverted location is necessary. For prolonged continuation of this situation, there may be a white/ black board at Control Room/ duty room of Technical Workers, where the diversion will be noted by the Operator/ SAE concerned for information. Proper communication with all related outsourced agency & Station Manager of concerned CCC must be ensured.

**Action to be taken by--**

Immediate Action by AE: Technical of Division/ Station Manager of CCC

**Proposed time frame--**

Implementation by 31<sup>st</sup> December'2018

**D. Brief Description of the unsafe condition:** Malfunctioning of 33KV DBCR/ TPGO Isolator at Substation & 11KV Isolator at DTR.

**Details of unsafe condition/ unsafe act observed:**

During operation of isolator some time all blades do not open/ close simultaneously due to mechanical alignment defect



**Steps for improvement:**

The problem to be sorted out after survey & subsequent rectification to be made with necessary adjustment/ replacement of contacts as & when necessary

**Action to be taken by:**

Survey work with estimate to be done by AE: Technical of the Division  
Procurement action with order for execution by Divisional Manager

**Proposed time frame:**

For survey & estimate: by 31<sup>st</sup> January'2019  
Procurement & Implementation by 30<sup>th</sup> April'2019

**E. Brief Description of the unsafe act: Missing of feeder identification & non availability of requisite number of Shut down/ Break down/ Feeder diversion board at some of the 33/11 KV Substations.**

**Details of unsafe act observed:**

At some substations, feeder is not marked in the front & back of Control Panel & switchyard; the operator may be confused during allowing Shutdown.

Similarly, due to absence of sufficient number of Shut down/ Break down/ Feeder diversion board, confusion may arise.

**Steps for improvement:**

Immediate action for distinctive feeder marking/ indication at both front & rear side of all Control Panels as well as at Switchyard & availability of requisite quantity of Shut down/ Break down/ Feeder diversion board to be taken.

**Action to be taken by:**

Immediate Action by AE: Technical & AM(HR&A) of Division

**Proposed time frame:**

Positively by 31<sup>st</sup> December'2018

**F. Brief Description of the unsafe act: Non-availability of switching instruction at Substation**

**Details of unsafe act observed:**

Due to non-availability of switching instructions as well as single line diagram at most of the Substations, possibility of wrong & improper shutdown cannot be ruled out.

**Steps for improvement:**

Switching Instruction & Single line diagram of the Substation alongwith 33KV & 11KV linkage with other/ adjacent Substations to be displayed at the Control room & the Operators should be instructed/ trained accordingly for proper shut down of feeders & PTRs.

**Action to be taken by:**

To be implemented by AE: Technical under guidance of Divisional Manager.



**Proposed time frame--**

Positively by 31<sup>st</sup> January'2019.

**G. Brief Description of the unsafe act: Safety Meeting at Division & Zonal Level.****Details of unsafe act observed:**

It is observed that from most of the Divisions & Zones, minutes of Safety Meetings are not received at Corporate Safety Cell.

**Steps for improvement:**

The Safety Meeting to be conducted at Divisions & Zones as per standing Circulars of WBSEDCL.

**Action to be taken by:**

HR&A Officer of the Division & Zone.

**Proposed time frame:**

Immediate Action.

**H. Brief Description of the unsafe act : Non availability/ non use of proper PPEs by Technical work force for HT & LT network****Details of unsafe act observed:**

- i. During surprise inspection, it was observed that some Agency do not arrange for PPEs or provide non-standard PPEs for their workers.
- ii. During surprise inspection, it was also observed that technical workers of some Agencies do not use PPEs during execution of work, even when those are available.

**Steps for improvement:**

Rigorous Surprise checking is required by all level, specifically from Division & CCC towards availability & proper usage of by the technical workers. In case of any deviation, proper action against the Agency/ worker, as the case may be, to be initiated as per standing norms.

**Action to be taken by:**

Action by AE: Tech/ AM(HR&A)/ SAE of Division/ Station Manager/ SAE of CCC & Safety Officer of Zone.

**Proposed time frame:**

Immediate Action

**I. Brief Description of the unsafe act : Improper/ insufficient documentation at Substation/ CCCs.****Details of unsafe act observed:**

1. At some Substation Permit to Work card is neither requisitioned nor issued properly by the concerned personnel for any work in HT network.



2. At some Substation shut down register is not available/ maintained properly
3. Similar practice persists in CCC regarding Field Memo.
4. Observed absence of possession of SCC by Supervisors & Workmen's permit for skilled workers and Identity/ Employment card of outsourced workforces as a whole.
5. Observed absence of Statutory document compliance as per Labour Law for Workers of most of Agencies performing Erection Job.
6. Observed signature of attendance of one worker by another/ presence of ghost worker.
7. Sometimes approved Duty roster & Attendance Register is not available at Control Room/ CCC for Technical workers.

**Steps for improvement:**

- i. Issuance of Permit to work for HT Network at Substation for any type of work involving Shutdown or not should be mandatorily introduced immediately.
- ii. Shut down register to be opened/ properly maintained to all substations immediately. After opening, recording of all shutdowns to be ensured.
- iii. Similar Action to be taken for Field Memo in CCC.
- iv. The particular Supervisor/ Skilled Worker not possessing SCC/ WP should be debarred from job till production of such document.
- v. Statutory document compliance as per Labour Law for Workers of Agencies performing Erection Job to be ensured.
- vi. Physical presence of workers during signature in attendance register to be ensured. Performing of mutual duty of both Operating & Maintenance personnel to be stopped.
- v. Duty Roster & Attendance Register must be available with Control Room/ SM of CCC for inspection at any point of time.

**Action to be taken by:**

Action by AE: Tech/ AM(HR&A) of Division/ Station Manager of CCC.

**Proposed time frame:**

Immediate Action.

**J. Brief Description of the unsafe act: Routine & Surprise visits not conducted at worksite during any execution for observance of SoP and availability & use of PPEs.**

**Details of unsafe act observed:**

It is observed that from the monthly compiled report of Zonal Safety Officers received at Corporate Safety Cell that most of the Site Offices are not conducting routine as well as surprise inspection.

**Steps for improvement:**

The Safety Inspection to be conducted by the Site Officers as per standing Circulars of WBSEDCL.

**Action to be taken by:**

Station Manager, AE: Tech & AM(HR&A), DM of Division, HR&A Officer & RM of Region and Zonal Safety Officer.

**Proposed time frame:**

Immediate Action



**CONTINUOUS MEASURES FOR GENERATION OF AWARENESS AMONG GENERAL PUBLIC AND TECHNICAL WORKFORCE & SAFETY TRAINING:**

**A. Safety Awareness Programme for Technical Workers.**

**Details observation:**

It is observed that Safety Awareness Programme for Technical Workers is not being conducted at most of the Sites in regular manner.

**Steps for improvement:**

Safety Awareness Programme to be conducted at Work Site as per standard practice of WBSEDCL.

**Action to be taken by:**

HR&A Officer of the Division, Region & Zone and Safety Officer of the Zone.

**Proposed time frame:**

At least one bi-monthly programme at each Division.

**B. Safety Awareness Programme for General Public.**

**Details observation:**

It is observed that Safety Awareness Programme for General Public is not being conducted at most of the Site in regular manner.

**Steps for improvement:**

Safety Awareness Programme to be conducted at strategic places as per standard practice of WBSEDCL.

**Action to be taken by:**

HR&A Officer of the Division, Region & Zone and Safety Officer of the Zone.

**Proposed time frame:**

At least one programme in every month at each Division.

**C. Safety Training Programme for Technical Workers.**

**Details observation:**

It is observed that Safety Training arranged by HRD Department at EETI, requisite number of participants are not provided by the Agencies, resulting in miserable participation & futile effort by WBSEDCL. Moreover, no technical worker from Erection Agencies are provided for the Trainings, though accidents are no less frequent for those workforce

**Steps for improvement:**

Regular Safety Training Programme for Technical Workers being conducted by HRD Department to make them safety conscious & to adopt safe work procedure during execution of work to be continued. HR&A Officer of the Divisions should take initiative to interact with the Agencies for sparing their workers for training mandatorily.



**Action to be taken by:**

HRD Department & Assistant Manager (HR&A) of the Division.

**Proposed time frame:**

In addition to Scheduled Training as per prevailing Training calendar, a special drive of at least one Safety Training to be imparted to those Technical Workers, who are yet to receive any Training so far by 31<sup>st</sup> August'2019.

#####  
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13.10.2018.